

Supplementary Materials for “Landauer Efficiency of Self-Modeling: An Operational Scale of Vitality”

This document collects the extended material omitted from the main text to keep it within the Entropy length corridor (8000–10000 words). The structure mirrors the main-text sections with an “S” prefix: each block addresses the same topic as its main-text counterpart, but in expanded form.

S2.1 Operationalizing I_{pred} : the full protocol

Measuring I_{pred} directly in the sense of Equation (1a) of the main text requires access to all internal degrees of freedom of the system and is technically impossible for biological, urban, and cosmic systems. The standardized operationalization fixes four elements for each class of system.

- (a) *Sensory channel x_t* — the observable input states. For a bacterium, the ligand concentration at the chemotactic receptors; for a city, the flow of resources through the infrastructure (energy, matter, information); for the biosphere, the flux of solar radiation and the geochemical flows.
- (b) *Internal channel s_t* — measurable summaries of the configuration M_t . For a bacterium, the methylation level of the adaptation system; for a city, administrative-infrastructure parameters (communication density, decision density, budget profile); for the biosphere, atmospheric composition and biomass pools.
- (c) *Choice of past/future windows* with the value of τ specified. For a bacterium, the characteristic response time, of the order of a few generations; for a city, a quarter or a year; for the biosphere, a geological eon.
- (d) *Mutual-information estimation method*: Kraskov–Stögbauer–Grassberger; plug-in with bias correction; neural-MI estimation via a critic.

The proxies are not the definition of I_{pred} ; they are its operationalization. The same discipline applies throughout physical measurement: temperature is measured not by directly counting molecular velocities but through a calibrated proxy — the expansion of mercury. A discrepancy between different proxies for one system is a signal to refine the proxies, not to abandon the concept.

Why predictive rather than stored information. Still, Sivak, Bell, and Crooks set the precedent for a thermodynamic link between predictive information and dissipation in their work on the thermodynamics of prediction (Still et al. 2012). In the stationary regime with optimal coding, the information I_{stored} that the system retains about its own past and the predictive information I_{pred} it holds coincide. In the non-stationary regime their difference (information-irrelevant memory, *nostalgia*) is diagnostically meaningful and is associated with excess dissipation. The present framework inherits Still’s precedent and adds a key requirement: both the numerator and the denominator of η_L refer to one and the same physical system — the model and the dissipation belong to X . This self-payment requirement separates η_L from the classical setting of the thermodynamics of prediction, where the model and the dissipator may be distinct circuits.

S2.4 Cohort normalization: extended discussion and the sigmoidal alternative

Extended discussion of reference cohorts

The specific reference cohorts for the six paradigm cases of Section 3 are fixed as follows.

Bacteria (§ 3.5): a cohort of prokaryotic chemotactic systems, $n \sim 200$ from BiGG/KEGG models (*E. coli*, *B. subtilis*, *V. cholerae*, and the like).

Cities (§ 3.6): 120–500 large cities from the Bettencourt dataset (Bettencourt et al. 2007).

Biosphere (§ 3.7): the only case for which percentile normalization is impossible owing to the absence of a cohort; in its place, normalization by a theoretical upper limit is used — the planetary net primary production $\text{NPP} \sim 10^{14} \text{ W}$ is taken as the T_v norm.

Large language model (§ 3.4): a cohort of the most recent generations of foundation models, $n \sim 50$ since 2020.

Stars (§ 3.1): a main-sequence cohort, $n \sim 10^4$ from the Hipparcos catalogue.

Crystal and hurricane (§§ 3.2–3.3): normalization by zero — the structural zero $T_v = 0$ or $C_v = 0$ is set not by position within a cohort but by a constructive failure of the corresponding capacity (absence of own dissipation paying for the maintenance; absence of held predictive information). Formally: $\tilde{T}_v(X) := 0$ is adopted by convention for X with $T_v(X) = 0$, and $\tilde{T}_v(X) := 1$ for X with $T_v(X) \geq \max_{\text{cohort}} T_v$; within the support of the reference cohort — Equation (2a) of the main text.

The sigmoidal alternative

An alternative to percentile normalization is sigmoidal normalization $\tilde{T}_v = T_v / (T_v + T_v^{\text{ref}})$, with an independently chosen reference value T_v^{ref} for each capacity. The sigmoidal procedure yields an absolute scale and applies to unique systems, but it requires justifying T_v^{ref} , C_v^{ref} , and S_v^{ref} from independent physical considerations. This work uses the percentile procedure and retains the sigmoidal one as an alternative for applications that require an absolute scale or that operate on unique systems.

S3.4 Counterfactual screening of the LLM: full component-by-component analysis

We apply the counterfactual procedure of § 2.2 of the main text to the concrete case of a large language model. The candidates for inclusion in the boundary are examined in sequence.

Parametric weights (static after training). Removal: the model does not respond; outright failure of the loop. They belong to the boundary of the generator, but not to the operating circuit at the moment of inference (the weights are not updated by the error).

KV-cache (runtime context memory). Removal: the model loses the conversation context; over a window $\tau \sim$ the session time, $\Delta(-dF_X/dt) > 0$. It belongs to the boundary on session timescales.

Runtime server (GPU + OS). Removal: computation is impossible. It belongs to the mechanism boundary, but the energy is supplied by the external power grid.

Owner corporation (decision-maker over training, deployment, payment). Removal: the model in its current form ceases to be updated (no new versions), but the current version keeps running on the residual infrastructure. The corporation is external relative to the current version of the model; it enters the boundary when one considers the evolution of the model lineage over time.

Counterfactual screening yields two different verdicts depending on the chosen window. On the short session scale, the LLM boundary = {weights + KV-cache + runtime server}: $\eta_L > 0$ through the numerator, but $\eta_L \rightarrow 0$ through the denominator, because the external power grid, not own dissipation within the boundary, pays for the computation. On the evolutionary scale (generations of models) the boundary expands to the corporation, and η_L is then assigned to the corporate system — with its own financial flows, errors, and survival — not to the model as an agent. Counterfactual screening therefore grants the LLM vital status on neither scale; it operationalizes the intuition on which the diagnosis of § 3.4 of the main text rests.

S3.5 Worked example *E. coli*: detailed sensitivity analysis

Full calculation parameters

Numerator. The sensory channel $x_t \in \{0, 1\}$ is the binarized sign of the temporal gradient of the aspartate concentration (positive/negative) at a step of $\delta t \sim 0.1$ s (the characteristic response time of a single receptor cluster). The internal channel $s_t \in \{0, 1, 2, 3, 4\}$ is the number of methyl groups on the Tar receptor (five discrete states of the adaptation system). The prediction window $\tau \approx 10$ s is the characteristic adaptation time of the methylation system. The model of Tu et al. 2008 (Tu et al. 2008; Shimizu et al. 2010) gives the transition probabilities $P(s_{t+\delta t} | s_t, x_t)$ analytically from the stationary distribution of the Markov process. Alternatively, $I(s_t; x_{t+\tau})$ is estimated numerically with the KSG (Kraskov–Stögbauer–Grassberger) estimator on simulated trajectories.

For a single receptor cluster within the window τ one obtains $I_{\text{pred}} \approx 1\text{--}10$ bits (Mehta and Schwab 2012; Cheong et al. 2011), an order of magnitude consistent with the information-theoretic analysis of biochemical signaling. A typical *E. coli* carries ~ 7000 Tar receptors organized into clusters, and integrating over all clusters yields $I_{\text{pred}} \sim 10^3\text{--}10^4$ bits per generation. This is an estimate for a single sensory module; including Tsr, Trg, Tap, and Aer scales I_{pred} approximately linearly with the number of sensors.

Denominator. The heat output of *E. coli* in the stationary growth phase is $\sim 1\text{--}3$ pW per cell (Liu et al. 2020). Over a generation $\tau_d \sim 30$ min $\approx 1.8 \cdot 10^3$ s, the total dissipation is $E_{\text{actual}}^{\text{total}} \sim 2 \cdot 10^{-9}$ J/cell. The exergetic efficiency of metabolism — the fraction of the free energy of glucose representable as an ATP equivalent — is estimated at $\eta_{\text{ex}} \sim 0.4\text{--}0.6$. The exergetic part physically able to pay for Landauer erasures is $E_{\text{actual}} = E_{\text{actual}}^{\text{total}} \cdot \eta_{\text{ex}} \sim 10^{-9}$ J/cell. At $T = 310$ K the minimal cost per bit is $k_B T \ln 2 \approx 3 \cdot 10^{-21}$ J, whence

$$N_{\text{max}} = \frac{E_{\text{actual}}}{k_B T \ln 2} \sim \frac{10^{-9}}{3 \cdot 10^{-21}} \sim 3 \cdot 10^{11} \text{ bits per cell.}$$

This is the first, *exergetic*, bound: the number of bits the cell could in principle erase if all its free energy went into logically irreversible information operations. In reality only a small fraction of the exergy — of order 10^{-3} (Mehta and Schwab 2012) — goes into irreversible information operations; the rest goes to biomass synthesis, the maintenance of ionic gradients, and thermal losses. The corresponding *computational* bound is $E_{\text{comp}} \sim 10^{-3} \cdot E_{\text{actual}} \sim 10^{-12}$ J/cell, and $N_{\text{max}}^{\text{comp}} = E_{\text{comp}} / (k_B T \ln 2) \sim 3 \cdot 10^8$ bits/cell.

Sensitivity analysis

Varying the window τ over 1–100 s shifts I_{pred} by ± 1 order of magnitude: on short windows the Markov model yields less information, and on long ones it saturates at the limit imposed by the adaptation memory.

The binning of x_t (binary versus 4-level) changes I_{pred} by $\sim 30\%$ — substantially less than the uncertainty in the number of receptor clusters.

Including all five receptor families (Tar, Tsr, Trg, Tap, Aer) scales I_{pred} approximately linearly with the number of sensors, leaving η_L within one order of magnitude.

The estimate of η_{ex} (0.4 versus 0.6) changes the denominator — and the resulting η_L — by a factor of ~ 1.5 .

The resulting conservative estimate, accounting for all sources of uncertainty: $\eta_L \in [10^{-9}, 10^{-7}]$ through the exergetic denominator and $\eta_L^{\text{comp}} \in [10^{-6}, 10^{-4}]$ through the computational denominator.

The bottleneck of the calculation

The bottleneck is estimating I_{pred} for the full sensory system: the number of receptor clusters varies across strains and conditions and is taken here from literature estimates, not from first principles. The precise value requires simultaneous MI measurements on all known sensory channels of a single population, and this defines the content of the accompanying experimental program. The full reproducible calculation with code is provided as part of the Supplementary Materials (Python sources, captured outputs, parameter documentation).

S4.4 Counterfactual screening of the thermostat: full component-by-component analysis

We apply the counterfactual procedure of § 2.2 to a bimetallic thermostat X with hysteresis $\Delta T_{\text{hyst}} = 1$ K, controlling a room with a thermal inertia $\tau_{\text{room}} \sim 30$ min. The candidates for inclusion in the boundary of X are enumerated in sequence.

Bimetallic strip (sensor + actuator). Removal: the loop is broken, no regulation at all. $\Delta(-dF_X/dt) > 0$ on timescales $< \tau_d$. It belongs to the boundary.

Relay contact group (energy gating). Removal: the same — the loop is broken. It belongs to the boundary.

Heating element (effector via the external power grid). Removal: the heat source disappears; the bimetallic strip and the relay remain, but the $-dF_X/dt$ of the remaining system tends to zero not through a failure of the loop but through the exhaustion of its own free energy (the bimetallic strip slowly cools to ambient). It does not belong to the decision loop and is an external component.

Power grid as a current source. Analogous to the heater: external.

Set-point operator (external operator). Removal: the set-point is fixed, regulation is preserved. External.

The boundary of the thermostat under counterfactual screening is {bimetallic strip, relay}. Within the boundary, E_{actual} is the elastic relaxation of the metal plus the dissipation of the relay, $\sim 10^{-6}$ W. The internal channel that holds predictive information about the room temperature is the deformation of the bimetallic strip, with $I_{\text{pred}} \leq \log_2(\Delta T_{\text{room}}/\Delta T_{\text{hyst}}) \sim 5$ bits. Crucially, this I_{pred} refers to the environment “the room,” not to the environment “thermostat + room + power supply.” The loop of decisions paid for by the system’s own decay is incomplete for the thermostat: the room is heated by the external power grid, not by the bimetallic strip’s own free energy. Counterfactual screening yields $\eta_L^{\text{thermostat}} > 0$ only for the trivial task of “maintaining the bimetallic strip’s own shape,” not for the control task of “regulating the room temperature.” The discrepancy between these two interpretations is exactly what Bruineberg et al. call the transition from a Pearl blanket to a Friston blanket: one and the same physical system admits several self-consistent boundaries, and the choice between them is a methodological, not an ontological, discipline.

S4.4a Rebutting the critiques of Andrews 2021 and Williams 2020

The well-known critiques of the FEP bear on η_L asymmetrically.

Andrews (Andrews 2021) — “The math is not the territory” — attacks the FEP for the non-derivability of empirical content from the formalism: without additional hypotheses, F can be constructed post hoc for any system. η_L rebuts this critique by anchoring the numerator and the denominator to measurable quantities — predictive information through the proxy C_v , exergy through thermodynamic measurements — so that the requirement of a concrete operationalization removes the freedom of post hoc construction.

Williams (Williams 2020) — “Predictive coding and thought” — attacks predictive coding as a neural mechanism, not the FEP as a principle; this critique is neutral to η_L , because the program does not assert predictive

coding as a mechanism but uses predictive information as a formal measure, methodologically independent of any specific neural architecture. The distinction is fundamental for a coherent defense against conflated critique: predictive coding (Rao and Ballard 1999; Clark 2013) is a neuroarchitectural hypothesis, whereas predictive information (Bialek et al. 2001) is an information-theoretic measure.

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